Inorganic chemistry

Laboratory Manual II

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Contens (number of experiment is the same like in Laboratory program and Czech laboratory manual “Návody pro laboratoře z anorganické chemie”, D.Sýkorová, L.Mastný)

4.1 Vanadium

The chemistry of vanadium(V) in aqueous solutions is complex and depends on pH of the solution. Here we use strongly acidic solutions in which vanadium is present in the form of VO_2^+ ion. The colour changes for the complete reduction sequence are as follows:

(+V)	**(+V) and (+IV)**	**(+IV)**	**+3**	**+2**
yellow	green	blue	green	lavender
VO_2^+	VO_2^+ & VO^{2+}	VO^{2+}	$[V(H_2O)_6]^{3+}$	$[V(H_2O)_6]^{2+}$

Sulphur dioxide reduces $V^{(+V)}$ to $V^{(+IV)}$ only; zinc in hydrochloric acid will reduce $V^{(+V)}$ step by step to $V^{2+.}$
The reduction equations:

(+V) → (+IV):
with aqueous SO_2:

$$2VO_2^+ + SO_3^{2-}(aq) + 2H^+(aq) \rightarrow 2VO^{2+}(aq) + SO_4^{2-}(aq) + H_2O(l)$$

with zinc in HCl:

$$2VO_2^+(aq) + 4H^+(aq) + Zn(s) \rightarrow 2VO^{2+}(aq) + Zn^{2+}(aq) + 2H_2O(l)$$

(+IV) → (+3)
with zinc in HCl:

$$2VO^{2+} + Zn(s) + 4H^+(aq) \rightarrow 2V^{3+}(aq) + Zn^{2+}(aq) + 2H_2O(l)$$

(+3) → (+2)
with zinc in HCl:

$$2V^{3+}(aq) + Zn(s) \rightarrow 2V^{2+}(aq) + Zn^{2+}(aq)$$

4.1.1 Reduction of vanadate

Material: $NaVO_3.2H_2O$, KOH, H_2SO_4, Na_2SO_3 ($Na_2S_2O_5$), Zn
Procedure: Use 1 % solution of $NaVO_3.2H_2O$. Add 1 ml of H_2SO_4 (0.5 $mol.l^{-1}$) and 3 g of Na_2SO_3 or $Na_2S_2O_5$ to 50 ml of $NaVO_3.2H_2O$ solution in the Erlenmeyer flask and heat it. What change of colour occurred? Boil the solution gently to remove SO_2, then cool it and add 3 g of Zn (granule). Close the mouth of the flask with a Bunsen valve. Shake with the flask occasionally and observe changes of colour.
To make the Bunsen valve, put a stopper with a glass tubing into the mouth of the flask. Then put a piece of rubber tubing on the end of the glass tubing, cut a vertical slit in the rubber tubing and put a glass plug in there. This valve acts in one direction, it prevents access of air while release of the gas from the flask is permitted.

4.1.2 Reactions of VO_3^- (VO_4^{3-}) – *(use 1 % solution of vanadate)*

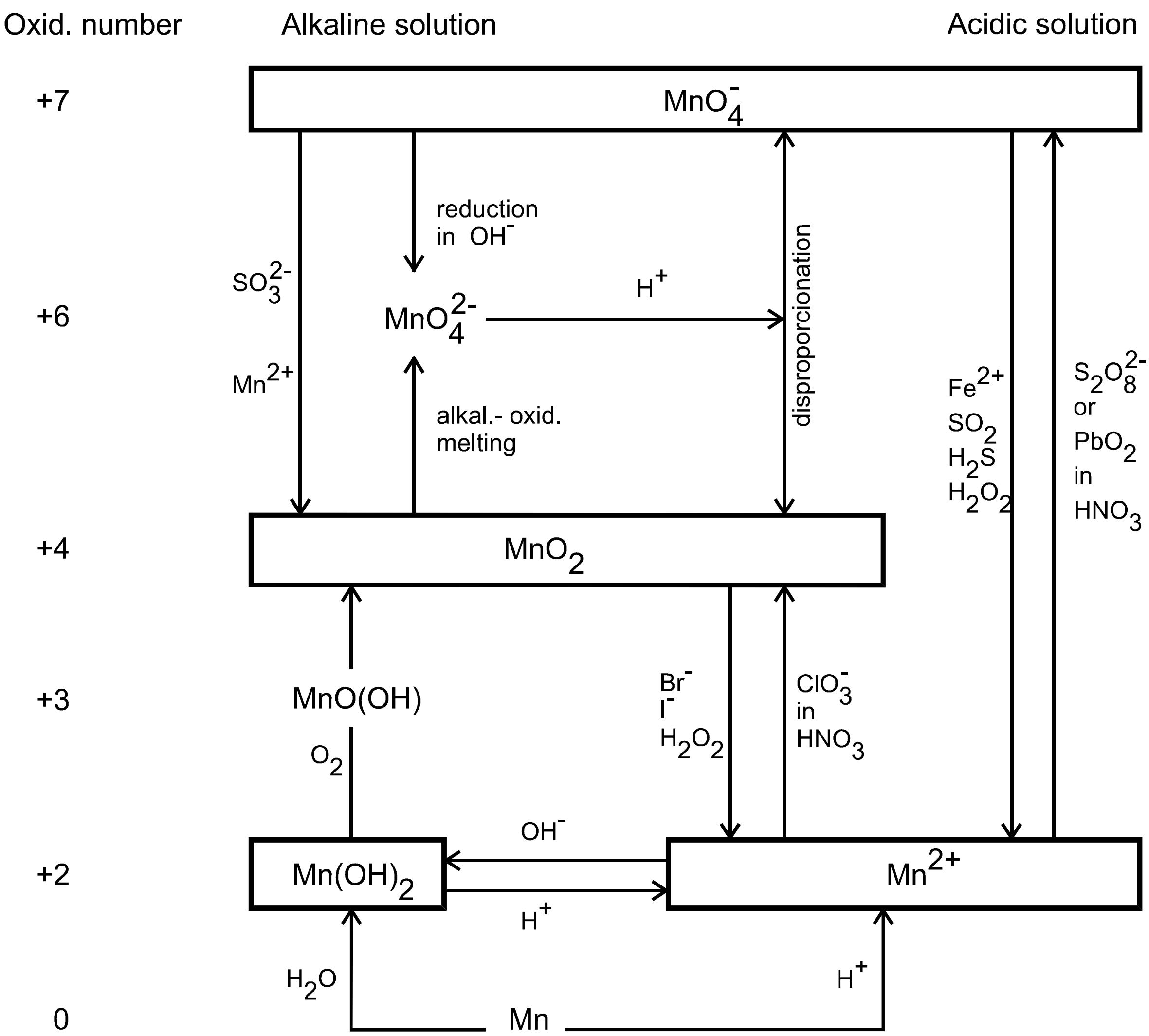
a) Add 1 ml of 20 % H_2SO_4 to 1 ml of VO_3^- solution, then add 3 % H_2O_2 solution (dropwise) until the colour changes. Then alkalize the solution with 20 % NaOH solution and observe change of colour.

b) Add 1 ml of concentrated HCl to 1 ml of VO_3^- solution in porcelain evaporation dish and evaporate (**in a fume hood**) to volume around 0.5 ml, observe change of colour.

c) Acidify 2-3 ml of VO_3^- solution with 20 % H_2SO_4 solution, add granule of Zn or Al and shake it. Observe colours of vanadium compounds during the reduction.

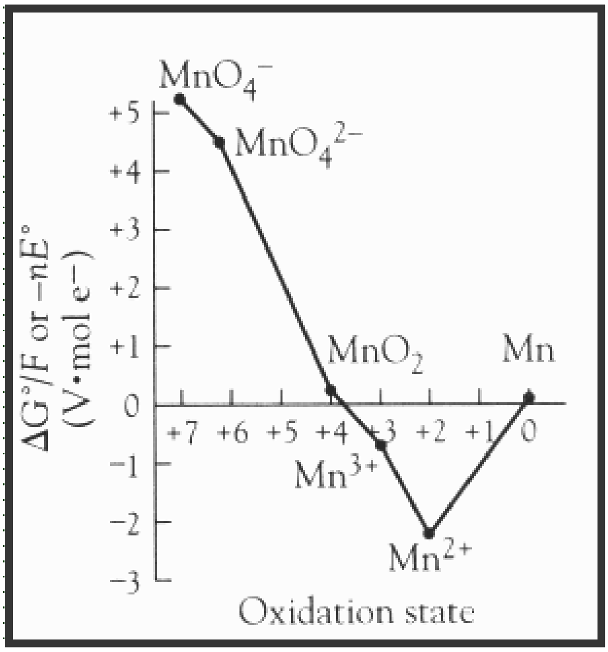
d) Add $(NH_4)_2S$ solution to the 2-3 ml of VO_3^- solution and observe colour of produced thiovanadate (VS_4^{3-}). Acidify the solution with diluted H_2SO_4 and observe colour of the precipitate (V_2S_5).

4.3. Manganese



Oxidation-reduction behaviour of manganese compounds

Chemistry of manganese from Frost oxidation state diagram:
This diagram plots the relative free energy of a species versus oxidation state. This diagram illustrates properties of different oxidation states of manganese.



The Frost diagram for manganese (pH = 0)

- Thermodynamic stability is found at the bottom of the diagram, the lower a species is positioned on the diagram, the more thermodynamically stable it is (from an oxidation-reduction perspective).
  Mn (II) is the most stable species.
- A species located on a concave curve can undergo disproportionation.
  MnO_4^{2-} and Mn (III) tends to disproportionate.
- Those species on a convex curve do not typically disproportionate.
  MnO_2 does not disproportionate.
- Any species located on the upper left side of the diagram will be a strong oxidizing agent.
  MnO_4^- is a strong oxidizer.
- Any species located on the upper right side of the diagram will be a reducing agent.
  M*anganese metal is a moderate reducing agent.*

4.3.1 Preparation of potassium permanganate (7 g) - fume hood, protecting shield, gloves

$$3\ MnO_2 + 6\ KOH + KClO_3 \rightarrow 3\ K_2MnO_4 + KCl + 3\ H_2O$$
$$3\ K_2MnO_4 + 4\ CO_2 + 2\ H_2O \rightarrow 2\ KMnO_4 + MnO_2 + 4\ KHCO_3$$

Material: MnO_2, KOH, $KClO_3$, CO_2 or CH_3COOH
Procedure: Mix crushed KOH (50 % excess) and $KClO_3$ (20 % excess) in a clean iron dish. While stirring with an iron rod, heat the mixture to 400 ^oC to make it melt. Stop heating when all the mixture melts. Add slowly powdered MnO_2 to the melted mixture. This procedure needs stirring with an iron rod. Heat the mixture until it solidifies.
Dissolve the solid mass in 300 ml of hot water and bubble CO_2 gas through the solution (it is possible to use diluted acetic acid instead). The reaction is finished

when a drop of the reaction mixture on the filter paper does not show presence of K_2MnO_4 (green colour).
Let the precipitate (MnO_2) settle down to bottom of the beaker and filter it with a fritted glass (do not use filter paper). Wash and dry the precipitate. Finally let the solution evaporate to crystallization and filter out the $KMnO_4$ crystals.

4.3.2 Reactions of MnO_4^- *(use 1 % $KMnO_4$ solution)*

a) Add concentrated solution of KOH (1-2 ml) to the same volume of $KMnO_4$ solution and heat it. Observe change of colour and write the equation of this reaction. Cool the solution and add diluted H_2SO_4 dropwise. What change in colour occurred?

b) Add 1 ml of $KMnO_4$ solutions to three test tubes and to each tube add 0.5 ml of one of the following solutions: H_2O, 10 % H_2SO_4, 10 % NaOH. Add five drops of diluted Na_2S solution to each test tube. Identify the reaction products by the colour and write the equations.

c) Add 1 ml of saturated $Ba(NO_3)_2$ solution to 1 ml of 1 % KI solution and a few drops of alkaline $KMnO_4$ solution. Observe a green precipitate of $BaMnO_4$ (Cassel green).

3.1.6 Preparation of barium nitrate

$$BaSO_4 + Na_2CO_3 \xrightarrow{t} BaCO_3 + Na_2SO_4$$
$$BaCO_3 + 2\ HNO_3 \rightarrow Ba(NO_3)_2 + CO_2 + H_2O$$

Material: $BaSO_4$, Na_2CO_3, HNO_3
Procedure: Pulverize 2 g of $BaSO_4$ with 10 g of Na_2CO_3. Put about 1 g of Na_2CO_3 at the bottom of a crucible than add the prepared mixture and further 1 g of Na_2CO_3 on the top. Heat the crucible (in a clay triangle) first gently and than intensively on the Mékere burner or in the electrical furnace at 900-1000 ^oC. Heat the melted mixture for about 20 minutes. Dissolve the solid mass in hot water. Filter the suspension, wash $BaCO_3$ with hot water to remove SO_4^{2-} ions. Dissolve $BaCO_3$ on the filter ~~in~~ with hot HNO_3 solution (1:1) – work in a **fume hood.** Evaporate the solution on a steam bath to the crystallisation and filter out the crystals. Precipitate the remaining Ba^{2+} ions in mother liquor in the form of $BaSO_4$.

$Ba(NO_3)_2$ – white crystals, soluble in water, split off oxygen during heating to form $Ba(NO_2)_2$ or even BaO when heated strongly.

3.2.4 Reactions of Al^{3+} *(use 0.2 mol.l^{-1} $Al_2(SO_4)_3$ or $KAl(SO_4)_2$ solution)*
(record your observations and write the balanced equations for all reactions)

a) Pour 2 ml of Al^{3+} solution to the four separated test tubes and to each tube add one of these reagents (dropwise): diluted NH_3, NaOH, Na_2CO_3 and Na_2S solutions. Compare the precipitates in all test tubes.
Dissolve the precipitate by adding a slight excess of NaOH solution, than add excess of NH_4Cl and heat the mixture.

b) Pour 2 ml of Al^{3+} solution to the four separated test tubes, add half a spoonful of tartaric acid and repeat the reactions (a). Compare the results.

c) Pour 2 ml of Al^{3+} solution to the two separated test tubes. Precipitate the solution in the first tube with 5 % Na_2HPO_4 solution. Repeat this reaction with hot solution of Al^{3+} in the second test tube. Compare the results.
Divide the precipitate into three test tubes and test its solubility in diluted HCl, CH_3COOH and NaOH.

d) Add 1-2 drops of diluted $Co(NO_3)_2$ solution to the 0.1-0.2 g of fast Al^{3+} compound in crucible. Anneal this mixture and observe the colour of $CoAl_2O_4$. For comparison that the colour is not only from Co^{2+} salt repeat this experiment with K_2SO_4 instead of Al^{3+} salt.

e) Add sufficient amount of NaOH solution to 1 ml of Al^{3+} solution to give $Na[Al(OH)_4]$ solution and add alizarin reagent. Then drop of 5 % acetic acid solution.

4.2.15 Reactions of Cr^{3+} *(use 0.3 mol.l^{-1} Cr^{3+} solution)*

a) Add NH_3(aq) dropwise to 2 ml of Cr^{3+} solution. Try the action of NH_3(aq) excess. Heat the mixture to the boil. Note and explain your observation.

b) Add 10 % NaOH solution (dropwise) to 2 ml of Cr^{3+} solution and than a few drops of H_2O_2. Observe the change of colour and write the reactions.

c) To 1 ml of Cr^{3+} solution add 1 ml of saturated $K_2S_2O_8$ solution and a few drops of 5 % $AgNO_3$ solution (catalyst). Shake 1-3 minutes and observe change of colour.

d) Add a few crystals of $KClO_3$ to the 1 ml of Cr^{3+} solution and 3 ml of concentrated HNO_3. Heat the mixture to boiling and observe change of colour.

e) Repeat reaction d) without $KClO_3$. Compare the results, explain the role of HNO_3.

f) Pour 1 ml of Cr^{3+} solution to the two separated test tubes. Add (dropwise) 10 % Na_2CO_3 solution to one and 10 % Na_2S solution to the other. What are the created compound and the released gases? Write the equations.

4.4.5 Reactions of Fe^{3+} *(use 0.5 mol.l^{-1} solution of Fe^{3+} salt)*

a) Put 1 ml of Fe^{3+} solution in each of the seven test tubes (or on a spot plate). Add a few drops of the following seven reagents: 1 % NH_4SCN, 5 % $K_3[Fe(CN)_6]$, 5 % $K_4[Fe(CN)_6]$, 5 % NaH_2PO_4, 5 % NaOH, 5 % KI and NH_3(aq) to each of the seven test tubes. Observe reactions and record them.

b) Add 5 % KF solution and 1 % NH_4SCN solution to 1 ml of Fe^{3+} solution. Explain your observation.

4.4.6 Reactions of Fe^{2+} *(use 0.5 mol.l^{-1} solution of fresh prepared Fe^{2+} salt)*

a) Repeat reactions 4.4.5 a) using Fe^{2+} solution in place of Fe^{3+} solution.

b) Add one drop of H_2SO_4 (1:2) and a few drops of $KMnO_4$ solution to 1 ml of Fe^{2+} solution. Then add a few drops of NH_4SCN solution.

The cyano complexes ("hexacyanoferrates") belong to the most stable amongst the iron complexes. The potassium salts are known as *"yellow prussiate of potash"* $K_4[Fe(CN)_6]$ or *"red prussiate of potash"* $K_3[Fe(CN)_6]$.
When a solution of potassium ferrocyanide is mixed with iron(III) chloride, or a solution of potassium ferricyanide is mixed with iron(II) chloride, both cases lead to the formation of the same colloidally *"dissolved Berlin blue"* $K[Fe^{III}Fe^{II}(CN)_6]$ at a molar ratio of 1:1.

$$K_4[Fe(CN)_6] + FeCl_3 \rightarrow K[Fe^{III}Fe^{II}(CN)_6] + 3\ KCl$$
$$K_3[Fe(CN)_6] + FeCl_2 \rightarrow K[Fe^{III}Fe^{II}(CN)_6] + 2\ KCl$$

Addition of excess iron(III) or iron(II) ions to $[Fe(CN)_6]^{4-}$ or $[Fe(CN)_6]^{3-}$, respectively, leads to the formation of blue precipitates, distinguished as *"insoluble Berlin blue"* or *"insoluble Turnbull's blue"*, respectively. However, they are chemically identical, $Fe_4[Fe(CN)_6](s)$.
The mixture of iron(III) chloride with red prussiate of potash, or iron(II) chloride with yellow prussiate of potash, however, results in clear or only weakly coloured solutions.

$$K_3[Fe(CN)_6] + FeCl_3 \rightarrow [Fe^{III}{}_2(CN)_6] + 3\ KCl \qquad \text{light brown}$$
$$K_4[Fe(CN)_6] + FeCl_2 \rightarrow K_2[Fe^{II}{}_2(CN)_6] + 2\ KCl \qquad \text{clear}$$

4.4.9 Reaction of Co^{2+} *(use 0.5 mol.l^{-1} Co^{2+} salt solution)*

a) Add 1 ml of saturated NH_4SCN solution to 1-2 ml of Co^{2+} solution (Vogel reaction). Observe colouration due to the formation of $[Co(SCN)_4]^{2-}$ complex and extract the product into 1 ml of ether.

b) To a solution of Co^{2+} slowly add concentrated HCl and observe change of colour. Dilute this solution with water. Note and explain the observation.

c) Add slowly concentrated NH_3 solution to 2 ml of Co^{2+} solution and observe what happens in excess of $NH_3(aq)$.

d) Carefully add 10 % NaOH solution to 2 ml of Co^{2+} solution. Observe colour of precipitate. Is the solid formed soluble in the excess of base? Divide the suspension and to one half add 3 % H_2O_2 solution. To the other half drop a few drops of bromine water. Write the equations.

e) Add drop of acetic acid and a few crystals of KNO_2 to 2 ml of Co^{2+} solution. Write the equation If you know that the yellow precipitate is $K_3[Co(NO_2)_6]$.

f) Write some lines on filter paper with help of glass rod soaked in Co^{2+} solution. After drying the lines miss. Carefully heat the paper. Explain this observation.

4.4.11 Reactions of Ni^{2+} *(use 0.5 mol.l^{-1} solution of Ni^{2+} salt, test tubes or a spot plate)*

a) Add diluted NH_3 and 0.5 ml of dimethylglyoxime reagent to 2 ml of Ni^{2+} solution. *The Ni^{2+} is first complexed with ammonia and then detected as an insoluble, scarlet coordination compound of dimethylglyoxime ($DMGH_2$).*

b) Add NaOH solution to 2 ml of Ni^{2+} solution to complete the precipitation. Divide the precipitate into two test tubes. Add excess of NaOH to one and a few drops of H_2O_2 to the other.

c) Add NH_3(aq) dropwise to 2 ml of Ni^{2+} solution. Try the action of NH_3(aq) excess. Then add NaOH solution. Record and explain your observations.

d) Add about 10 drops of 2 % NaOH solution and a small amount of solid $Ca(ClO)_2$ to 2 ml of Co^{2+} solution. Write the equation. Decant the solid in test tube with water (two times) and divide it into two parts for reactions e) and f).

e) Add 1 ml of 2 % Fe^{2+} solution, one drop of concentrated HCl and one drop of 5 % KSCN solution to the first part of solid from d). Observe the change of colour and write the equation.

f) To the second part of solid from d) add slowly 3 % H_2O_2 solution. Write equation of this reaction.

g) Add concentrated HCl to a small amount of fast Ni^{2+} compound to dissolve it. Dilute this solution with water (observe change of colour) and add 20 % Na_2CO_3 solution. Write the formulas of all three compounds created in these reactions.

4.5.11 Reactions of Cu^{2+} *(use 0.3 mol.l^{-1} solution of Cu^{2+} salt, test tubes or a spot plate)*

a) Add about 0.5 ml of $K_4[Fe(CN)_6]$ solution to 1-2 ml of Cu^{2+} solution. Try the solubility of precipitate in HCl, CH_3COOH and NH_3(aq).

b) Repeat the reaction (a) with 5 % $K_3[Fe(CN)_6]$ solution.

c) Add 10 % NaOH solution to 1-2 ml of Cu^{2+} solution to complete the precipitation. Heat the mixture and observe change of colour.

d) Add 2 ml of tartaric acid solution to 1-2 ml of Cu^{2+} solution and repeat the experiment (c). Explain the difference.

e) Add concentrated NH_3(aq) (a few drops at a time) to 1-2 ml of Cu^{2+} solution. Record your observation. Add ammonia solution until there is no further change. Then add two pellets of NaOH and boil.

f) Add a few drops of 5 % KI solution to 1-2 ml of Cu^{2+} solution. Add a few drops of Na_2SO_3 solution to determine colour of the precipitate.

Precipitation of CuI, which is white, but the solution is brown because of formation of tri-iodide ions (iodine). After adding Na_2SO_3 to the solution, tri-iodide is reduced to colourless iodide ions and the white colour of the precipitate becomes visible.

g) Add concentrated HCl dropwise to 1-2 ml of Cu^{2+} solution and observe change of colour. Dilute the solution with water.

4.3.3 Reactions of Mn^{2+} *(use 0.3 mol.l^{-1} solution of Mn^{2+} salt)*

a) Add NH_3(aq) to 2 ml of Mn^{2+} solution. Observe reaction of precipitate with excess of ammonia. Then add 1-2 drops of H_2O_2.

b) Use very diluted Mn^{2+} solution (0.02 mol.l^{-1}), acidify it with HNO_3 and add solid $K_2S_2O_8$ and one drop of 1 % $AgNO_3$ solution. Heat the mixture. Compare with reaction without $AgNO_3$.

c) Add $KMnO_4$ solution dropwise to 1 ml of Mn^{2+} solution and observe changes of colour. Measure the pH value of the solution and write the equation of this reaction.

d) Add 0.5 ml of concentrated HNO_3 and a few drops of Mn^{2+} solution to a small amount of solid PbO_2 in test tube. Heat carefully to the boiling and then dilute with water (two times). Allow the precipitate to settle and observe colour of the solution. Write the equation.

e) Add a few drops of 5 % $AgNO_3$ solution and diluted NH_3 solution to a 1 ml of Mn^{2+} solution. Observe colour of product and write the equation.
The spot test on filter paper – to 1 drop of Mn^{2+} solution on filter paper add 1 drop of $AgNO_3$ solution and 2 drops of diluted NH_3 solution.

Ion-exchange chromatography

The ion-exchange chromatography process allows separation of ions and polar molecules according to the charge properties of the molecules. It is a special case of column chromatography. An interchange of ions occurs between a liquid mobile phase and an insoluble solid stationary phase that is in contact with the solution.
There are two general forms of resin (ionex, stationary phase): cationic and anionic.

Cationic exchange resins contain acidic groups ($-SO_3H$, $-PO(OH)_2$, -COOH).
Cation exchange is illustrated by the equilibrium:

$$n(RSO_3^-H^+) + M^{n+} \rightarrow (RSO_3^-)_nM^{n+} + nH^+$$

Ion-exchange selectivity (bonding to the stationary phase of the column) depends on the affinity of the resin for the cation (or anion) in the solution. The affinity depends on the size and charge of the solvated ion. The greater is the charge/size ratio of the ion passing through the column, the greater is its affinity for the resin.

$Pb^{2+}>Sr^{2+}>Ca^{2+}>Mg^{2+}>Rb^+>Na^+>Li^+$

$Fe^{3+}>Al^{3+}>Mg^{2+}>Na^{+}$

Anionic resins contain alkaline groups (quaternary ammonium or amine groups). Anion exchange is illustrated by the equilibrium:

$$n(RNH_4^{+})(OH^{-}) + A^{n-} \rightarrow (RNH_4^{+})_nA^{n-} + nOH^{-}$$

The affinity of anions depends on strength and proticity of acids.
$I^{-} > Br^{-} > Cl^{-} > F^{-} > OH^{-}$
$SO_4^{2-} > SO_3^{2-} > HPO_4^{2-} > NO_3^{-}$

Basic terms:

capacity: the total amount of positive charged ions that a column can exchange, in equivalents or milliequivalents to unit volume of the resin

charging: the process in which ions from a salt solution are adsorbed onto the column

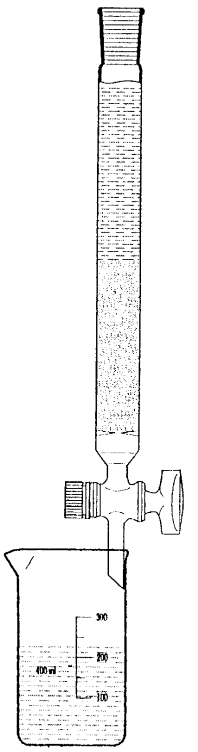
effluent: the solution emerging from the bottom of a column

eluant: the salt solution that is poured through the column to carry out an elution

elution: the process of de-adsorbing one kind of ion from the resin and replacing it with another kind of ion

ionic form: the identity of the ions that are originally adsorbed onto a column before use. The resin used in this experiment is supplied in the hydrogen form

regeneration: the process of returning a column to its original ionic form after use, regeneration is carried out by eluting the column with a concentrated solution of the desired ion.



Chromatography column

9.8.1 Ion exchange separation of cations

Preparation of the chromatographic column:
- fill the column with aqueous suspension of cation exchange resin (about 50 ml). Make sure that the column is tightly packed.
- convert the cation exchange resin to the H^+Res^- form by washing with 100 ml of HCl (2 mol/l).
- pour distilled water through the column in amount twice the column volume before checking pH of the eluent with litmus paper. Continue washing the column with distilled water until the litmus shows no positive test for H^+.

During this experiment, do not allow the level of liquid to fall below the top of settled resin.

Procedure:
a) Pour the solution of Co^{2+} onto the cation exchange column. Make sure that the drip rate in the column is slow enough to ensure equilibration. A rapid drip rate will not allow quantitative displacement of H^+ by the Co^{2+} ion.
When all the salt solution is spent pour a small volume of pure water through the column. Elute Co^{2+} with solution of HCl (6 mol/l). When the eluate from the column becomes coloured, begin collecting it. Propose a procedure for determination of the Co^{2+} concentration in the original solution.
Wash the column with water to pH neutral.

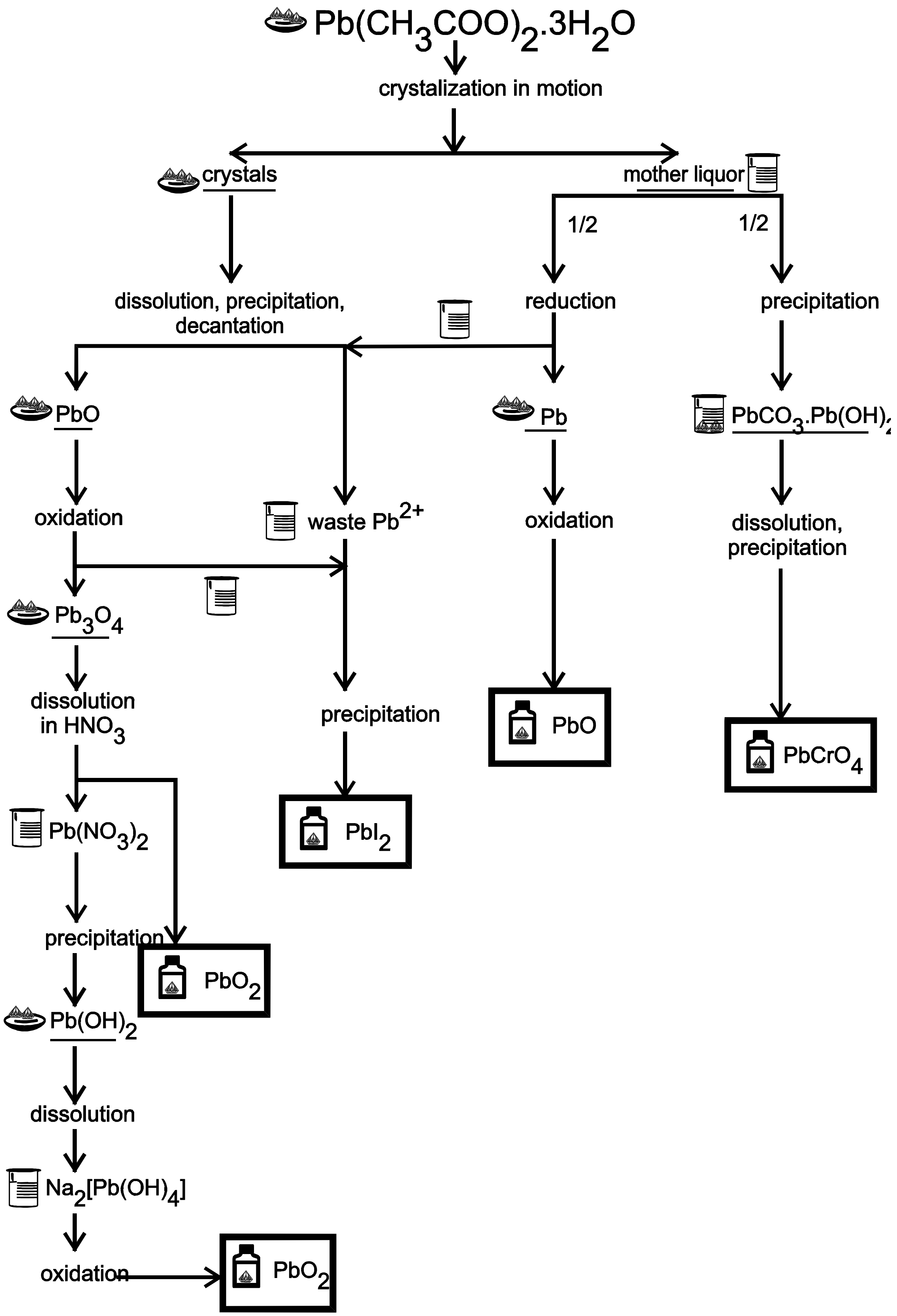
b) Mixture of Cu^{2+} and Co^{2+} cations in solution of 80 vol.% acetone – 20 vol.% HCl (250 ml of solution contains 5 g $CuCl_2.2H_2O$ and 10 g $CoCl_2.6H_2O$). Use procedure a) for separation of these cations.

4.8 Preparation cycles

These preparations start with a pure chemical substance and subject it to a series of chemical reactions. Several important types of reactions (oxidation-reduction, acid-base, precipitation, temperature decomposition, creation of complexes) and techniques (crystallization, decantation, melting, drying, sintering, work with gases) of synthesis and separation are involved. The qualitative reactions of some anions and cations are used to control purity of the products.

These reaction series are wasteless, all secondary products are utilized. In the end of those experiments you will calculate element balance in all products and work out the laboratory report.

Preparation cycle A



3.3.2 Preparation of lead

a) $Pb(CH_3COO)_2 + Zn \rightarrow Pb + Zn(CH_3COO)_2$

Material: $Pb(CH_3COO)_2.3H_2O$, Zn (pellets), CH_3COOH
Procedure: Dissolve 5 g of $Pb(CH_3COO)_2.3H_2O$ to solution where w($Pb(CH_3COO)_2$) = 0.1 and add 1-2 ml of CH_3COOH. Add calculated amount of Zn (+ 10 % excess) and let the mixture stay for 2-3 hours. Remove Pb from Zn pellets during this time.
Test for the end of reduction: add a few drops of H_2SO_4 to 1 ml of the solution in a test tube. Wash the formed lead with water acidified with acetic acid, then with pure water and dry it.

b) $PbO + C \rightarrow Pb + CO$

Material: PbO, charcoal
Procedure: Crush 4 g of PbO in a mortar and dry it at about 110 ^oC. Heat the charcoal in a closed crucible and pulverize it after cooling. Mix dry PbO with charcoal (20 % excess to stoichiometry) and put this mixture into a crucible. Heat the crucible first gently and then strongly in oxidizing flame of the Mékere burner for about 20 minutes. Empty the content of crucible on the tile and let it cool. Separate Pb from Zn (small pieces by floating with water).

3.3.3 Preparation of lead carbonate

$$Pb(CH_3COO)_2 + (NH_4)_2CO_3 \rightarrow PbCO_3 + 2\ CH_3COONH_4$$
$$Pb(NO_3)_2 + 2\ NaHCO_3 \rightarrow PbCO_3 + 2\ NaNO_3 + CO_2 + H_2O$$

Material: $Pb(CH_3COO)_2$ or $Pb(NO_3)_2$, $(NH_4)_2CO_3$ or $NaHCO_3$
Procedure: Add saturated $(NH_4)_2CO_3$ or $NaHCO_3$ solution (10 % excess to stoichiometry) to saturated Pb^{2+} salt solution while continuously stirring. Decant the precipitated product with water, filter it and dry at laboratory temperature.

$PbCO_3$ – white powder, insoluble in water, easy soluble in acids and hydroxides, decomposes ~~by~~ on heating. If Na_2CO_3 is used for precipitation, the alkali carbonate $2PbCO_3.Pb(OH)_2$ creates.

3.3.4 Preparation of lead oxide (thermal decomposition of $Pb(OH)_2$, $Pb(NO_3)_2$ or $PbCO_3$)

$$PbCO_3 \rightarrow PbO + CO_2$$
$$Pb(NO_3)_2 \rightarrow PbO + 2\ NO_2 + \frac{1}{2}\ O_2$$
$$Pb(OH)_2 \rightarrow PbO + H_2O$$

Material: $Pb(OH)_2$
Procedure: Add finely powdered $Pb(OH)_2$ to hot solution of 35 % KOH (7 g KOH for 1 g of $Pb(OH)_2$). Stir and heat the mixture to dissolve the $Pb(OH)_2$. Let the solution cool, the crystallization of PbO occurs. Filter out the crystals, wash them with water and dry at 105 ^oC.

PbO – yellow (orthorhombic), red (tetragonal) crystals, insoluble in water.

3.3.5 Preparation of lead dioxide

a) $Pb(NO_3)_2 + 2\ NaOH \rightarrow Pb(OH)_2 + 2\ NaNO_3$
$Pb(OH)_2 + 2\ NaOH \rightarrow Na_2[Pb(OH)_4]$
$Na_2[Pb(OH)_4] + CaOCl_2 \rightarrow PbO_2 + CaCl_2 + 2\ NaOH + H_2O$

Material: $Pb(NO_3)_2$ or $Pb(CH_3COO)_2$, NaOH, $CaOCl_2$ or NaClO, HNO_3
Procedure: Dissolve 0.1 mol of Pb^{2+} salt in 300 ml of hot water and cool the solution. If a small amount of substance remains undissolved, it will not affect the result. Add a solution of 20 g NaOH in 180 ml of water. First white precipitate of $Pb(OH)_2$ appears, which dissolves in excess of NaOH to $Na_2[Pb(OH)_4]$.

Mix 40 g of $CaOCl_2$ with 50 ml of water and the necessary amount of Na_2CO_3 for reaction with Ca^{2+}. Add 200 ml of water to the mixture and filter it.
Add the filtrate to the boiling solution of $Na_2[Pb(OH)_4]$ until the test for presence of Pb^{2+} in the final solution is negative.

Test for presence of Pb^{2+}: add one drop of Na_2S solution to one drop of the reaction mixture on filter paper. Black precipitate (PbS) indicates presence of Pb^{2+}.

Pour the mixture into 500 ml of water and decant it. Add 100 ml of diluted HNO_3 (1:3) to the precipitate, stir and decant it until pH is neutral. Filter off the lead dioxide, wash it with boiling water and dry.

b) $Pb(NO_3)_2 + 2NaOH \rightarrow Pb(OH)_2 + 2NaNO_3$
$Pb(OH)_2 + 2NaOH + Cl_2 \rightarrow PbO_2 + 2NaCl + 2H_2O$

Material: $Pb(NO_3)_2$ or $Pb(CH_3COO)_2$, NaOH, Cl_2
Procedure: Dissolve 10.0 g of $Pb(NO_3)_2$ in 80 ml of water acidified with a drop of concentrated nitric acid and add solution of 2.4 g of sodium hydroxide dissolved in 50 ml of water slightly with constant stirring. Heat the precipitated lead(II) hydroxide suspension to 70 - 80 ^oC and bubble chlorine through it at the same time (do not heat it over 80 ^oC or PbO will be obtained!). Decant the precipitated brown-black product with diluted nitric acid (1:3) and then with water. Dry it at 100 ^oC.

PbO_2 – brown powder, insoluble in water, decomposes on heating to Pb_3O_4 or PbO and oxygen. Good oxidizer.

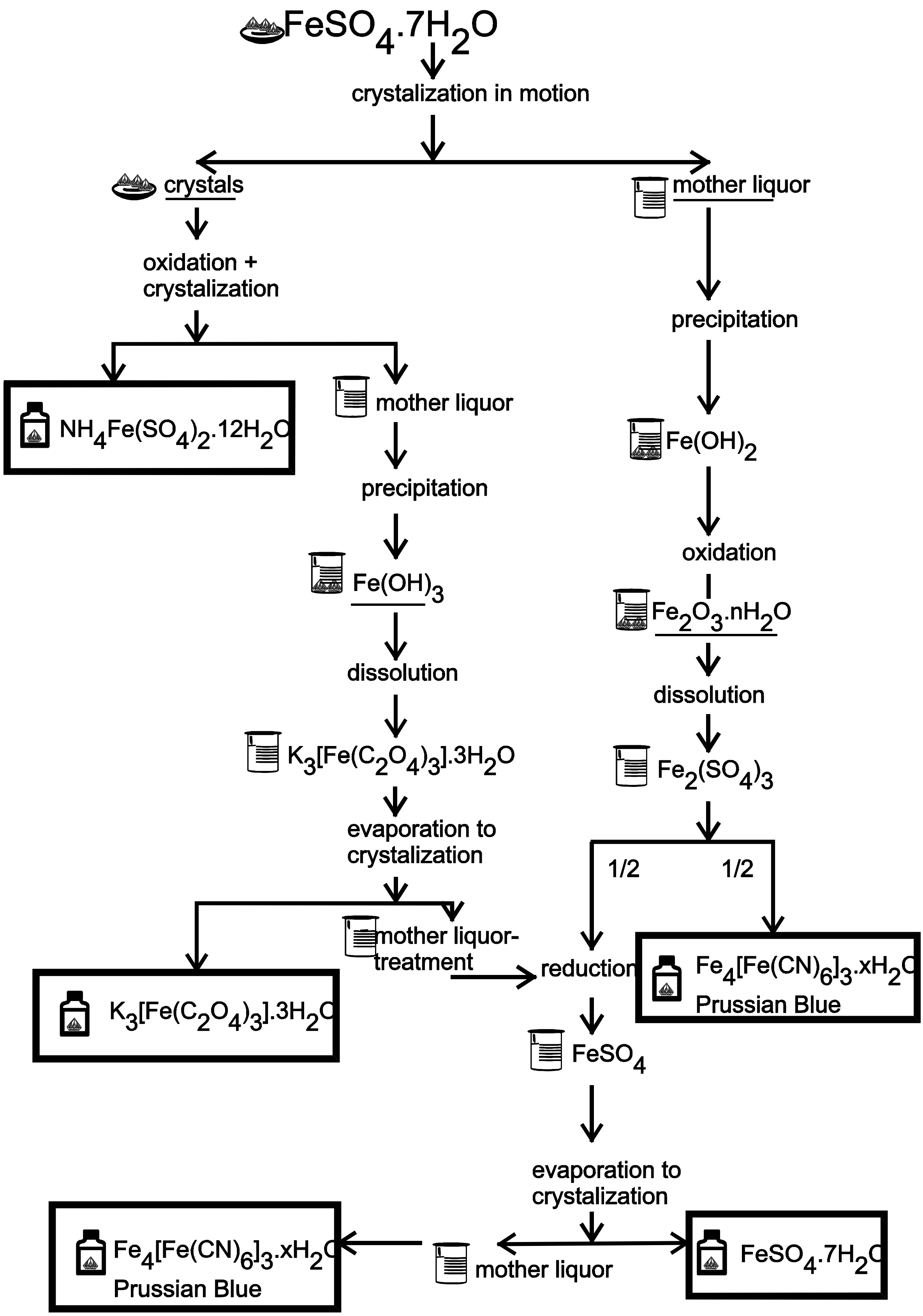
3.3.6 Preparation of Pb_3O_4 -“red lead” (5 g)

$$3\ PbCO_3 + NaNO_3 \rightarrow Pb_3O_4 + NaNO_2 + 3\ CO_2$$

Material: $PbCO_3$ (PbO), $NaNO_3$, $KClO_3$
Procedure: Mix the calculated amounts of $PbCO_3$ and $NaNO_3$ well together. Stir and heat it gently in a porcelain crucible. First decomposition of $PbCO_3$ to yellow PbO and CO_2 occurs at 200 ^OC. When all mixture is yellow increase the temperature to about 400 - 450 ^OC (crucible is not red) and stir continually. When the mixture turns red add 1-2 g $KClO_3$, stir and heat it for 30 minutes. If the temperature is higher (470 ^oC) Pb_3O_4 decomposes.

Pb_3O_4 (2PbO.PbO_2, minium, red lead) – red powder, good oxidizer, at 500 ℃ decomposes to PbO and oxygen.

Preparation cycle B



4.4.3 Preparation of potassium tris(oxalate)ferrate(III) trihydrate

$$Fe(OH)_3 + 3\ KHC_2O_4 \rightarrow K_3[Fe(C_2O_4)_3] + 3\ H_2O$$

Material: $FeSO_4.7H_2O$ or $(NH_4)_2Fe(SO_4)_2.4H_2O$, $K_2C_2O_4$, $H_2C_2O_4$, HNO_3, ethanol
Procedure: Dissolve 35 g of $FeSO_4.7H_2O$ in 100 ml of warm water and add slowly diluted HNO_3 (1:1) to oxidize Fe^{2+}. Add $NH_3(aq)$ to the solution until the precipitation of $Fe(OH)_3$ is completed. Let the precipitate settle and decant the liquid. Filter out the precipitate and wash it with hot water. Prepare a hot solution of 44 g of KHC_2O_4 (calculate it as a mixture of $K_2C_2O_4$ and $H_2C_2O_4$) in 100 ml H_2O. Add precipitate of $Fe(OH)_3$ in small portions to this solution. Filter the resulting solution and evaporate it on a steam bath to crystallization. Filter out and wash the crystals on the Buchner funnel with ethanol/water 1:1 and finally with acetone. Transfer the product to a dry filter paper and let it dry in air.

$K_3[Fe(C_2O_4)_3].3H_2O$ – green crystals, photosensitive and decomposes due to influence of light:

$$2\ K_3[Fe(C_2O_4)_3] \rightarrow K_2[Fe(C_2O_4)_2] + K_2C_2O_4 + 2\ CO_2$$

4.4.4 Preparation of iron alum

$$2\ FeSO_4 + H_2O_2 + H_2SO_4 \rightarrow Fe_2(SO_4)_3$$
$$Fe_2(SO_4)_3 + (NH_4)_2SO_4 \rightarrow 2\ NH_4Fe(SO_4)_2$$

Material: $FeSO_4.7H_2O$, H_2O_2, H_2SO_4, $(NH_4)_2SO_4$
Procedure: Dissolve $FeSO_4.7H_2O$ in water to a solution with $w(FeSO_4) = 0.15$. Filter this solution if necessary and carefully add concentrated H_2SO_4 in 10 % excess to the stoichiometry. Then slowly add H_2O_2 (double amount compared to stoichiometry), while stirring the mixture continuously. Heat the mixture to the boiling. Make sure that Fe^{2+} was oxidized to Fe^{3+} by a reaction of sample with $K_3[Fe(CN)_6]$. Add further H_2O_2 if necessary.
Evaporate the solution on a steam bath to half the volume and add warm saturated (by 60 ^oC) solution of calculated amount of $(NH_4)_2SO_4$. Let the solution crystallize. Put the crystals on a dry filter paper and let them dry in air.

$NH_4Fe(SO_4)_2.12H_2O$ – colourless or light violet crystals, turn brown on air.

Preparation of Prussian blue - $Fe_4[Fe(CN)_6]_3 \bullet 14H_2O$

$$Fe^{3+} + 3\ [Fe(CN)_6]^{4-} \rightarrow Fe_4[Fe(CN)_6]_3$$

Material: Fe^{3+} salt, $K_4[Fe(CN)_6]$
Procedure: Dissolve iron(III) salt in water in a beaker and $K_4[Fe(CN)_6]$ in another one. Slowly add $K_4[Fe(CN)_6]$ solution to the iron(III) solution, while stirring. Let the blue precipitate settle. Decant the supernatant liquid and wash the precipitate with water. Swirl the mixture and let the precipitate settle, decant and repeat the washing once more. Filter out the product and let it dry by air pulling air through the filter for a few minutes.

If $K_4[Fe(CN)_6]$ is added in excess to Fe^{3+} salt, a product with the composition of $KFe[Fe(CN)_6]$ is formed. This tends to form colloidal solutions (soluble Prussian blue) and cannot be filtered.

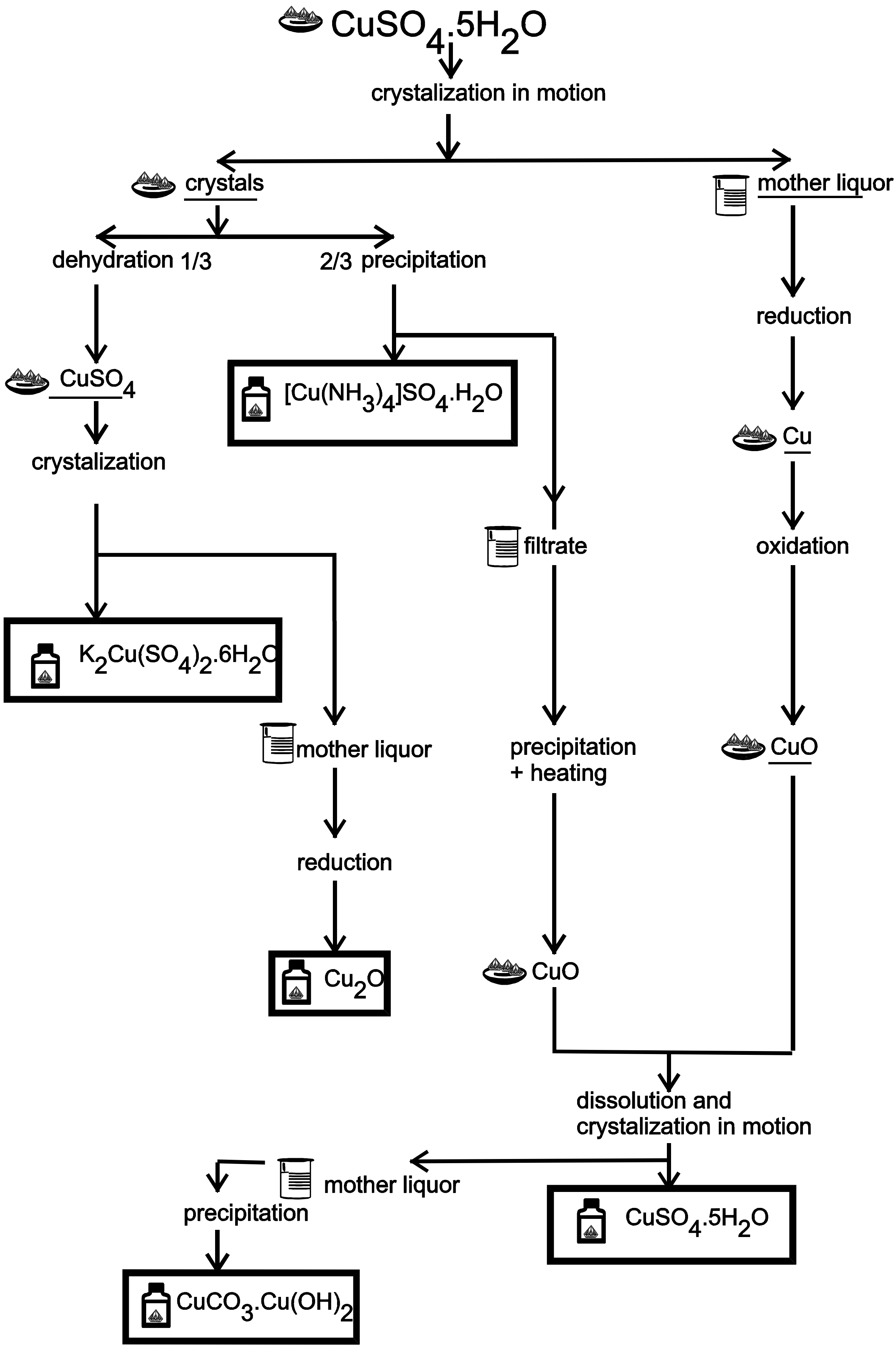
4.4.8 Preparation of sodium hexanitrito cobaltate (III) (7 g)

$$Co(NO_3)_2 + 7\ NaNO_2 + 2\ CH_3COOH \rightarrow Na_3[Co(NO_2)_6] + NO + 2\ NaNO_3 + \\ + 2\ CH_3COONa + H_2O$$

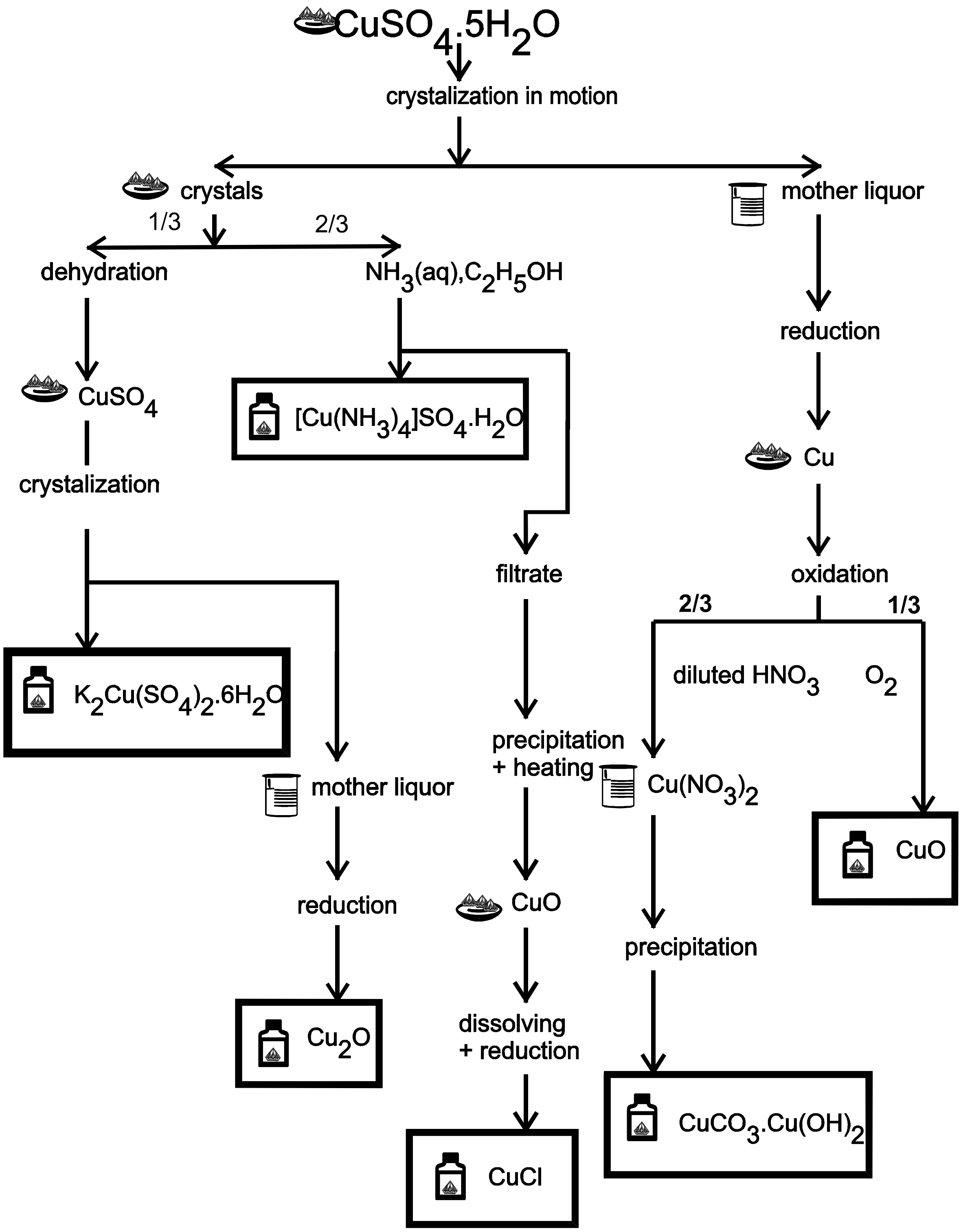
Material: $Co(NO_3)_2.6H_2O$, $NaNO_2$, CH_3COOH, C_2H_5OH
Procedure: Calculate the required amounts of parent compounds for preparation of 7g of $Na_3[Co(NO_2)_6]$. .Dissolve $NaNO_2$ in the equal weight of warm water. Cool this solution to 45–50 ^oC and add solid $Co(NO_3)_2.6H_2O$ while stirring continuously. Transfer the solution to the boiling flask and add 50 % CH_3COOH solution in small portions, swirling the flask slowly (work in a **fume hood**). After half an hour transfer the mixture to the washing bottle (or filter flask with glass tube). Suck air through the mixture with help of water aspirator (for half an hour). Connect the bottle to a water trap with NaOH. Filter the mixture and add slowly ethanol to the solution until crystals form. Filter off the product, wash it with small amount of ethanol and dry it on air.

$Na_3[Co(NO_2)_6].0.5H_2O$ – yellow micro-crystals, soluble in water, used for confirmation of K^+ ions.

Preparation cycle C



Preparation cycle D



4.5.2 Preparation of copper

$$CuSO_4 + Fe(Zn) \rightarrow Cu + Fe(Zn)SO_4$$

Material: $CuSO_4.5H_2O$, Fe (nails) or Zn, H_2SO_4, HCl
Procedure: Prepare 10 % solution from 4 g $CuSO_4.5H_2O$ and add a few drops of concentrated H_2SO_4. Place the beaker with solution on wire gauze, heat it gently to 50 ^oC and add iron nails or Zn (in 20 % excess). Heat and stir it until the colour of solution turns light green. Filter off the copper.

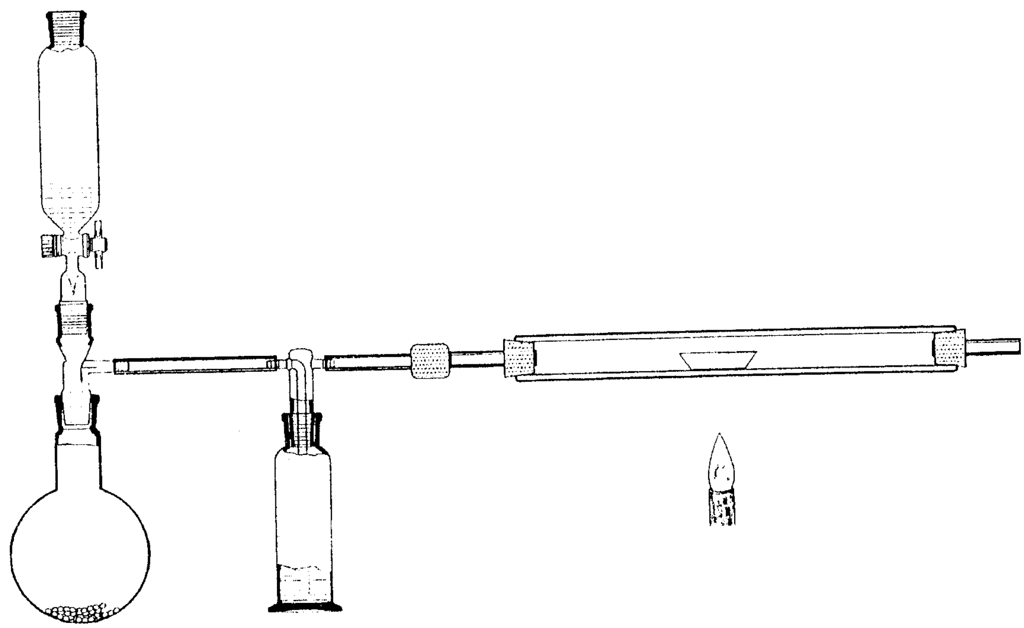
The excess of Zn will be dissolved from the product by diluted HCl, which does not affect the copper. Add in small portions 10 ml of HCl (1:1) in about 5 minutes. Heat it gently and stir to determine if hydrogen is still evolving from the excess of Zn used. Filter out the copper and wash it with water. Store the copper under the water.

4.5.3 Preparation of copper oxide

$$2\ Cu + O_2 \rightarrow 2\ CuO$$

Material: Cu (from experiment 4.5.2), for oxygen preparation: H_2O_2, MnO_2
Procedure: Set up a gas production apparatus and connect it to combustion tube (see figure)**,** where a ceramic boat with Cu is placed. Pass the oxygen through the wash bottle with concentrated H_2SO_4 and heat the tube under the boat for 20-30 minutes. Discontinue the heating, but continue passing oxygen through the apparatus until the combustion tube is sufficiently cool to be touched by hand.

CuO – back powder, insoluble in water, soluble in acids.



Apparatus for oxidation (reduction) by gas

4.5.4 Preparation of copper(I) oxide - cuprous oxide

$$2\ CuSO_4 + 5\ NaOH + HOCH_2(CHOH)_4COH \rightarrow$$
$$\rightarrow\ Cu_2O + HOCH_2(CHOH)_4COONa + 2\ Na_2SO_4 + 3\ H_2O$$

Material: $CuSO_4.5H_2O$, potassium sodium tartarate – $KNa.C_4H_4O_6.4H_2O$, glucose, NaOH
Procedure: Prepare a solution of 18 g $CuSO_4.5H_2O$ in 100 ml water and add another solution which contains 30 g of potassium sodium tartarate and 10 g NaOH in 100 ml of water. Add a solution of 8 g glucose in 300 ml water and boil the final solution for 1-2 minutes. Filter off the red precipitate of Cu_2O, wash it to remove SO_4^{2-} ions and alkaline reaction of the filtrate. Wash the product with ethanol and dry it on air.
Heat a small sample of Cu_2O in a crucible and observe change of colour, what is the product?

Cu_2O – yellow or red powder (the colour can be varied by changing the particle size), insoluble in water. Formation of Cu_2O oxide is the basis of the sensitive Fehling's test for sugars.

4.5.6 Preparation of copper ammine sulphate - perform this experiment in a fume hood

$$CuSO_4 + 4\ NH_3(aq) + H_2O \rightarrow [Cu(NH_3)_4]SO_4.H_2O$$

Material: $CuSO_4.5H_2O$, $NH_3(aq)$, C_2H_5OH
Procedure: Place 5 g of finely powdered copper sulphate, $CuSO_4.5H_2O$, in a small beaker, pour upon it 7.5 ml of concentrated ammonia and 3 ml of water. Shake it for about 1 minute and then heat it gently until all the solid dissolves. Add about 10 ml of ethanol to the solution, let it stand for about one hour and filter off the crystals. Wash them with a mixture of 5 ml of concentrated ammonia and 5 ml of ethanol. Dry them on air in the hood.

$[Cu(NH_3)_4]SO_4.H_2O$ – dark blue crystals, soluble in water (18 g in 100 ml of water at 21.5 ^oC), stable on air.

4.5.8 Preparation of cuprous chloride – work in a fume hood

a) $Cu^{2+} + Cu + 2\ Cl^- \rightarrow 2\ CuCl$

Material: $CuSO_4.5H_2O$, Cu, NaCl, HCl, Na_2SO_3, CH_3COOH
Procedure: Prepare a solution of 10 g of powdered $CuSO_4.5H_2O$ and 15 ml of concentrated HCl in a 250-ml round-bottom flask, add 4 g of NaCl and heat to boiling. Cover the flask with a little funnel. Add copper to hot solution in small portions. The green colour of the solution will turn to yellow. Filter off the remaining Cu. Pour the solution to one litre of cold water with 2 g of Na_2SO_3.
Wash the precipitated cuprous chloride 2 – 3 times with a solution of 1.5 g of Na_2SO_3, 6 ml of HCl and 300 ml of H_2O by decantation. Filter out the precipitate and wash it with concentrated acetic acid. Dry it in a drying oven at 100 ^oC.

b) $2\ Cu^{2+} + SO_2 + 2\ Cl^- + 2\ H_2O \rightarrow 2\ CuCl + SO_4^{2-} + 4\ H^+$

Material: $CuSO_4.5H_2O$, NaCl, HCl, SO_2, CH_3COOH
Procedure: Add 5 g of NaCl to the warm solution (70 ^oC) of 10 g $CuSO_4.5H_2O$ and bubble SO_2 through the mixture. CuCl precipitates. Filter out the precipitate and wash it with a solution of SO_2 in water and then with concentrated acetic acid.

CuCl – white crystals, insoluble in water, on air turns to green alkali copper chloride.

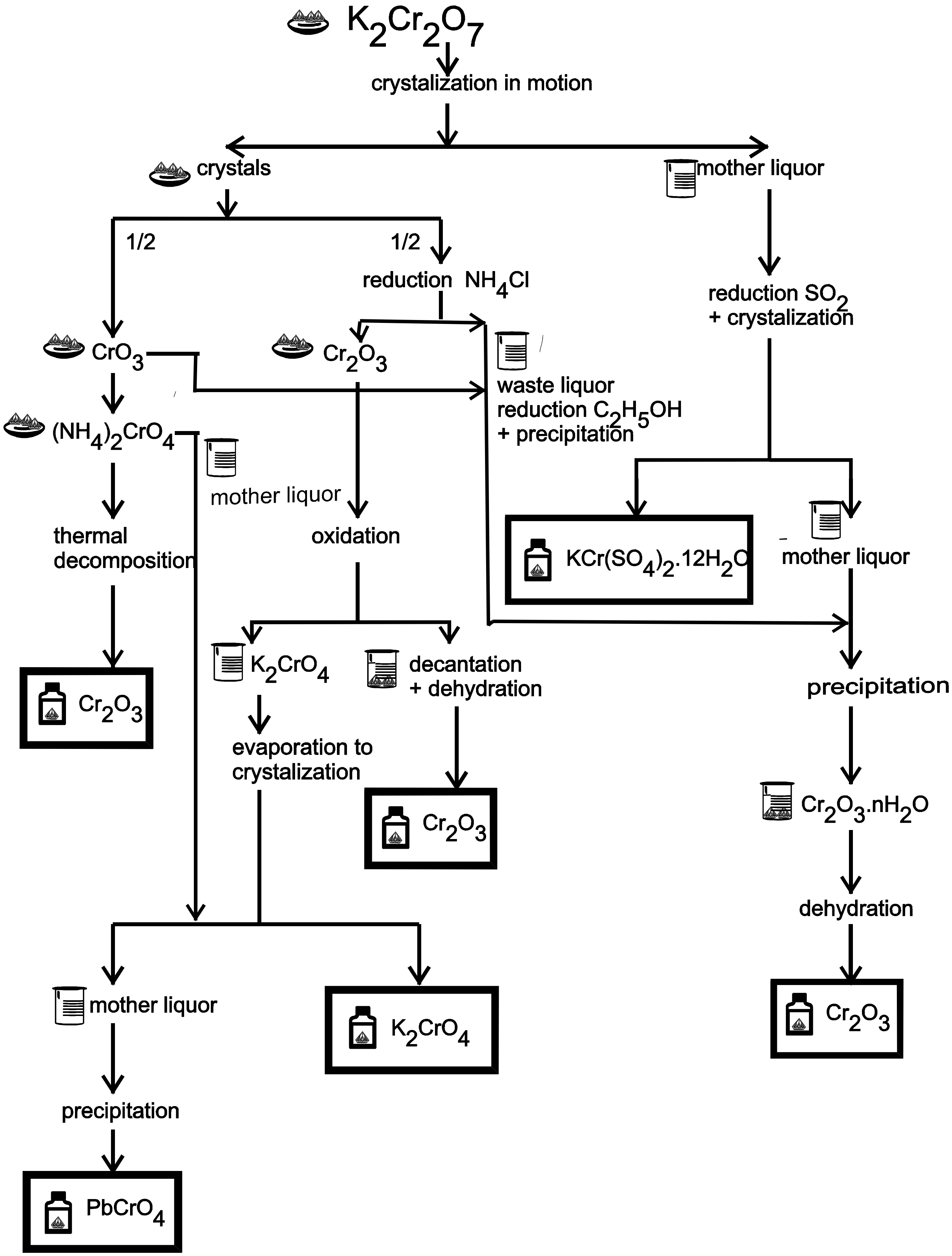
4.5.9 Preparation of basic copper carbonate

$$2\ Cu(NO_3)_2 + 2\ Na_2CO_3 + H_2O \rightarrow CuCO_3.Cu(OH)_2 + CO_2 + 4\ NaNO_3$$

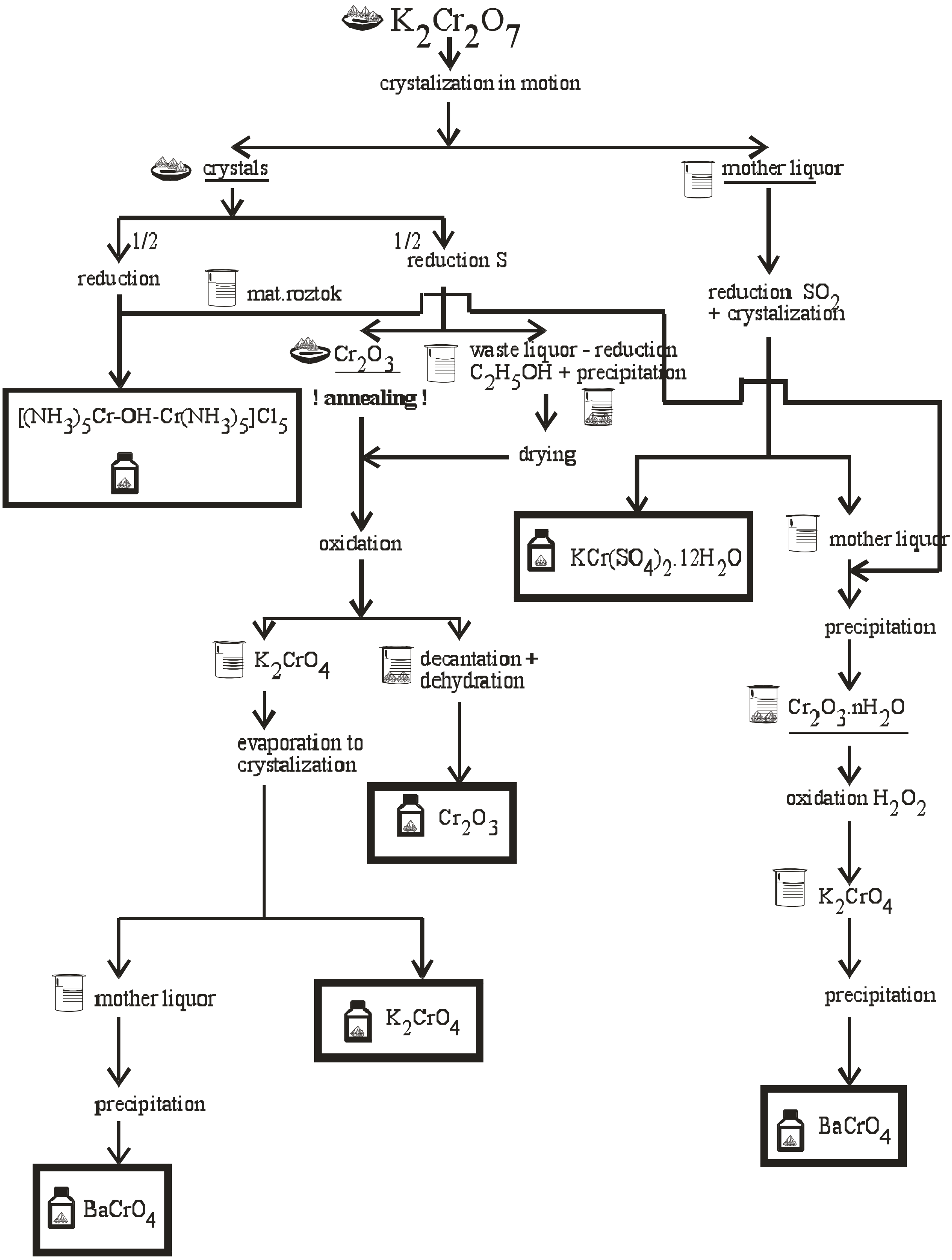
Material: Cu^{2+} salt, Na_2CO_3
Procedure: Add stoichiometric amount of Na_2CO_3 solution to the aqueous solution of Cu^{2+} salt, a little at a time with stirring, until all the fizzing is stopped and the reaction is complete. Let the mixture stay (for several hours) to change the precipitate into crystalline form. Filter out the product, wash it with water to remove anions of Cu^{2+} salt and let it dry on the filter paper.

$CuCO_3.Cu(OH)_2$ (malachite) – green crystals, insoluble in water, soluble in NH_3(aq).

Preparation cycle G



Preparation cycle H



4.2.1 Preparation of chromium(III) oxide – work in a **fume hood**

a) $K_2Cr_2O_7 + 2\ NH_4Cl \rightarrow Cr_2O_3 + N_2 + 2\ KCl + 4\ H_2O$

Material: $K_2Cr_2O_7$, NH_4Cl
Procedure: Place a mixture of 10 g of pulverized $K_2Cr_2O_7$ and NH_4Cl (2.5 times more than stoichiometric amount) in a crucible. Heat the crucible first gently and then intensively for about an hour. Let the mixture cool and dissolve the solid mass in 250 ml of boiling water. Filter out the chromium oxide. The filtrate is colourless if the reaction was made properly. Treat the orange filtrate to obtain $Cr_2O_3.xH_2O$ (see 4.2.3). Wash chromium oxide with water until the reaction on Cl^- ion is negative and dry it in a drying oven.

b) $K_2Cr_2O_7 + S \rightarrow Cr_2O_3 + K_2SO_4$

Material: $K_2Cr_2O_7$, S
Procedure: Prepare a mixture of 10 g of $K_2Cr_2O_7$ and S in 30 % excess to the stoichiometry. Heat this mixture in the iron plate and observe change of colour. Let the black-green product cool and remove it from the plate. Pulverize the solid mass, heat it with 250 ml of water and filter out the chromium oxide. The filtrate should be colourless. Treat the orange filtrate to obtain $Cr_2O_3.xH_2O$ (see 4.2.3). Wash chromium oxide with water until the reaction on SO_4^{2-} ion is negative and dry it in a drying oven.

Cr_2O_3 – green powder (it is black when prepared by the high temperature process), insoluble in water.

4.2.3 Preparation of chrome alum - $KCr(SO_4)_2.12H_2O$

a) $K_2Cr_2O_7 + 3\ SO_2 + H_2SO_4 \rightarrow KCr(SO_4)_2 + H_2O$

Material: $K_2Cr_2O_7$, SO_2, H_2SO_4
Procedure: Blow the SO_2 gas through a gas washing bottle with $K_2Cr_2O_7$ solution acidified with H_2SO_4. Do not permit the temperature to rise above 60 ^oC. Above this temperature complexes of chromium (III) sulphate are formed. These complexes contain sulphate in a non-ionisable form and are difficult to crystallise.
Bubble the unreacted gas through a washing bottle with 10 % NaOH solution.

Test for the end of the reaction – to the sample of reduced solution in a test tube add small amount of Na_2CO_3 crystals and heat the mixture just below the boiling point. Let the precipitate settle, the solution over the precipitate has to be colourless.
Set the solution aside to crystallise after the end of reduction. When the crystallisation is complete, filter off the crystals and wash them with a small amount of water. Transfer the product to a dry filter paper and let them dry in air.

b) $K_2Cr_2O_7 + 3\ C_2H_5OH + 4\ H_2SO_4 \rightarrow KCr(SO_4)_2 + 3\ CH_3CHO + 7\ H_2O$

Material: $K_2Cr_2O_7$, C_2H_5OH, H_2SO_4

Procedure: Dissolve crushed $K_2Cr_2O_7$ in diluted H_2SO_4 (1:3) and add, in small portions with stirring, calculated volume (+ 10 % excess) of C_2H_5OH. Do not permit the temperature to rise above 60℃. Continue like i n procedure a).

$KCr(SO_4)_2.12H_2O$ – dark violet crystals, crystallize in regular octahedra, soluble in water.

4.2.5 Preparation of chromium(VI) oxide

$$K_2Cr_2O_7 + 2\ H_2SO_4 \rightarrow 2\ CrO_3 + 2\ KHSO_4 + H_2O$$

Material: $K_2Cr_2O_7$, H_2SO_4, HNO_3
Procedure: Crush up 10 g of $K_2Cr_2O_7$ with a mortar and pestle and dissolve it in 25 ml of water. Slowly and carefully pour concentrated H_2SO_4 (4-times excess to stoichiometry) to this solution. Let the mixture stay for 1-1.5 hour to crystallize $KHSO_4$. After the crystallization filter out $KHSO_4$ using a fritted glass. Heat the filtrate to about 85 ^oC and add in small portions concentrated H_2SO_4 (about 1/2 of the previously used amount). Let the solution crystallize. After the crystallization filter out CrO_3 using a fritted glass. Wash the crystals with concentrated HNO_3 and dry them on the air. Treat the mother liquor to obtain $Cr_2O_3.xH_2O$!

CrO_3 – red strongly hygroscopic crystals, soluble in water, the substance is a strong oxidant and reacts violently with combustible and reducing materials causing fire and explosion hazard. The solution in water is a strong acid, it reacts violently with bases and is corrosive. The substance is very toxic to aquatic organisms. It is strongly advised that this substance does not enter the environment.

4.2.7 Preparation of ammonium chromate

Material: CrO_3, NH_3(aq)
Procedure: Dissolve CrO_3 (not necessarily dry) in diluted NH_3 solution. Cool the mixture in an ice bath. Filter the solution and evaporate it on the steam bath to crystallization. Let the solution crystallize. Filter out the crystals and precipitate insoluble chromate (see 4.2.10) from the mother liquor.

$(NH_4)_2CrO_4$ – yellow crystals, soluble in water.

4.2.10 Preparation of lead chromate and barium chromate (precipitation)

$$K_2Cr_2O_7 + 2\ NH_3 + H_2O \rightarrow K_2CrO_4 + (NH_4)_2CrO_4$$
$$K_2CrO_4 + Pb(NO_3)_2 \rightarrow PbCrO_4 + 2\ KNO_3$$
$$(NH_4)_2CrO_4 + Pb(NO_3)_2 \rightarrow PbCrO_4 + 2\ NH_4NO_3$$

For preparation of $BaCrO_4$ use $Ba(NO_3)_2$ or $BaCl_2$ instead of $Pb(NO_3)_2$.

Material: $K_2Cr_2O_7$, K_2CrO_4, $Pb(NO_3)_2$, NH_3(aq)

Procedure: Calculate the amount of reagent for precipitation of Cr(VI) in your sample. Weigh it and prepare 5 % solution. Use the 5 % solution of $K_2Cr_2O_7$ or K_2CrO_4, too. Add 10 % NH_3 solution to the $K_2Cr_2O_7$ solution before precipitation until the pH is moderately alkaline.
Mix 5 % K_2CrO_4 solution and 5 % solution of the precipitating agent (Pb^{2+} or Ba^{2+} salt) while continuously stirring (keep 10 volume% of each solution for finishing the precipitation). The end of precipitation can be tested by the following way: put 2 drops of the solution cleared up above the precipitate on a glass plate. Add a drop of the precipitating agent to one drop and a drop of CrO_4^{2-} solution to the other one. If a yellow precipitate appears in one drop or another, add approx 1 ml of the appropriate solution to the reaction mixture and repeat the test.
Decant the precipitate, filtrate it and wash with water to remove foreign ions. Dry the pure product.

$PbCrO_4$, $BaCrO_4$ – yellow powders, insoluble in water and acetic acid.

4.2.14 Preparation of µ-hydroxo-bis(pentaamminechromIII) – work in a fume hood

$$2\ CrCl_2 + 9\ NH_3 + NH_4Cl + \tfrac{1}{2}\ O_2 \rightarrow [(NH_3)_5Cr\text{-}OH\text{-}Cr(NH_3)_5]Cl_5$$

Material: $K_2Cr_2O_7$, HCl, C_2H_5OH, NH_4Cl, $NH_3(aq)$
Procedure: Spread 12 g of $K_2Cr_2O_7$ and mix it with 7.2 ml of C_2H_5OH and 28.2 ml of concentrated HCl in a flask. Write equation of this reaction.
Put the calculated amount of Zn (for reduction of Cr^{3+} to Cr^{2+} and 50 % excess) into a 250-ml flask and add $CrCl_3$ solution. Close the flask with a stopper with two holes. One is for a separatory funnel with 15 ml of concentrated HCl, the second is for a glass tube. Observe change of colour from green (Cr^{3+}) to blue (Cr^{2+}) after adding HCl. Pour the solution to the flask with a mixture of 250 ml $NH_3(aq)$ and 100 g of NH_4Cl. Swirl the flask and observe change of colour to red. Red crystals should precipitate in a few minutes. Filter off the crystals and wash them with diluted HCl (1:2). Recrystallize the crude product by dissolving in small volume of water and precipitate with diluted HCl (1:2). Filter off the pure product, wash it with a small amount of ethanol and dry it on air.

$[(NH_3)_5Cr\text{-}OH\text{-}Cr(NH_3)_5]Cl_5$ – red crystals, soluble in water. Creates blue complex $[(NH_3)_5Cr\text{-}O\text{-}Cr(NH_3)_5]Cl_4$ in NH_3 or NaOH solutions.